How to Make Water Safe in an Emergency

SEPT. 19, 2024, VISIT LINK FOR DETAILS.

ESPAÑOL

For Everyone

KEY POINTS

- After an emergency, your tap water may not be available or safe to use.
- Use bottled, boiled, or treated water to avoid getting sick.
- Boiling is the best way to kill germs in water.

MORE INFORMATION



Overview

After an emergency or disaster—such as a water main break, hurricane, or flood—you can take steps to make sure you have safe water.

If you know or suspect your water is unsafe, **do not** use that water to:

- Drink
- Wash dishes
- Brush your teeth
- Wash or prepare food
- Wash your hands
- Make ice
- Make baby formula

Instead, use bottled, boiled, or treated water for drinking, cooking, and personal hygiene. Follow recommendations from your health department for boiling or treating your water.

Also consider if there are other sources of safe water in or near your home you could use. Never use water from radiators or boilers that are part of your home heating system for drinking, cooking, or hygiene. This water could make you sick.

Water with fuel or toxic chemicals in it

You **cannot** make water containing fuel, toxic chemicals, or radioactive materials safe by boiling or disinfecting it. If you know or suspect your water

contains fuel or toxic chemicals: 1) Use bottled water or a different source of water 2) Contact your health department for specific advice.

Boil

If you don't have safe bottled water, boil your water to make it safe to drink.



Boiling is the best way to kill germs in water.

Steps for boiling water

If the water is cloudy, first filter it through a clean cloth, paper towel, or coffee filter. Another option is to let it settle. Then, draw off the clear water and follow the steps below.

1. Bring clear water to a rolling boil for 1 minute (at elevations above 6,500 feet, boil for 3 minutes).
2. Let the boiled water cool.
3. Store the boiled water in clean, sanitized containers with tight covers.

Making boiled water taste better

Improve the flat taste of boiled water by:

- Pouring it from one container to another and then letting it stand for a few hours
- Adding a pinch of salt for each quart or liter of boiled water

Disinfect

You can kill most germs in water with chemical disinfectants, such as unscented household chlorine bleach, iodine, or chlorine dioxide tablets. However, disinfectants do not work as well as boiling for killing some germs, including the parasites *Cryptosporidium* and *Giardia*.

Steps for using bleach to disinfect water

If the water is cloudy, first filter it through a clean cloth, paper towel, or coffee filter. Another option is to let it settle. Then, draw off the clear water and follow the steps below.

1. Follow the instructions on the bleach label for disinfecting drinking water. If the label does not have instructions, check the "active ingredient" on the label to find the bleach's concentration of sodium hypochlorite. In the United States, it will typically be between 5% and 9% sodium hypochlorite. Then, based on the concentration you have, use the tables below to add the appropriate amount of bleach to the water.
2. Stir the bleach and water mixture well.
3. Let the water stand for at least 30 minutes before you drink it.
4. Store the disinfected water in clean, sanitized containers with tight covers.



See how much bleach to add to different amounts of water in the tables below.

Bleach with a 5% to 9% concentration of sodium hypochlorite

Use the measurements below to make water safe to use. If the water is cloudy, murky, colored, or very cold, add double the amount of bleach listed below.

1 quart or liter of water

- If you have a dropper: Add 2 drops of bleach
- If you have something that measures in milliliters (mL): Add 0.1 mL of bleach
- If you have a measuring spoon: Add a tiny amount (too small to measure)

1 gallon of water

- If you have a dropper: Add 8 drops of bleach
- If you have something that measures milliliters (mL): Add ½ mL of bleach
- If you have a measuring spoon: Add a little less than ⅛ teaspoon

5 gallons of water

- If you have a dropper: Add 40 drops of bleach
- If you have something that measures milliliters (mL): Add 2½ mL of bleach

- If you have a measuring spoon: Add ½ teaspoon of bleach

Bleach with a 1% concentration of sodium hypochlorite

Use the measurements below to make water safe to use. If the water is cloudy, murky, colored, or very cold, add double the amount of bleach listed below.

1 quart or liter of water

- If you have a dropper: Add 10 drops of bleach
- If you have something that measures milliliters (mL): Add ½ mL of bleach
- If you have a measuring spoon: Add ⅛ teaspoon of bleach

1 gallon of water

- If you have a dropper: Add 40 drops of bleach
- If you have something that measures milliliters (mL): Add 2½ mL of bleach
- If you have a measuring spoon: Add ½ teaspoon of bleach

5 gallons of water

- If you have a dropper: Add 200 drops of bleach

- If you have something that measures milliliters (mL): Add 12½ mL of bleach
- If you have a measuring spoon: Add 2½ teaspoons of bleach

Using chemical tablets or drops to disinfect water

Follow the instructions on the label or package of chlorine dioxide, chlorine, or iodine tablets or drops. Chemical disinfectants are made to disinfect specific amounts of water.



Chemical disinfectants kill viruses and bacteria, but may not kill parasites.

If *Cryptosporidium* is in your water

Iodine, tablets with iodine (tetraglycine hydroperiodide), or chlorine tablets kill most germs, but will **not** kill *Cryptosporidium* germs. Chlorine dioxide tablets will kill *Cryptosporidium* as long as you follow the instructions correctly.

Who should avoid water disinfected with iodine

If you are pregnant, have thyroid problems, or have a hypersensitivity to iodine, avoid drinking water disinfected with iodine. Additionally, no one should drink water disinfected with iodine for more than a few weeks at a time.

Filter

Portable water filters do not remove all types of germs from water.

Portable filters:

- Will not remove viruses
- Must have an absolute pore (hole) size of 0.3 micron or smaller to remove bacteria
- Must have an absolute pore size of 1 micron or smaller to remove parasites (such as *Giardia* or *Cryptosporidium*)

Using a filter

1. Follow the instructions for the water filter you are using.
2. After filtering, add a chemical disinfectant—such as iodine,

chlorine, or chlorine dioxide—to the filtered water. The disinfectant will kill any viruses and bacteria.



Most filters will remove parasites, but not viruses or bacteria.

Using light

UV light

Portable units that deliver ultraviolet (UV) light can kill germs in small amounts of clear water. UV light does not work well in cloudy water because small particles can block germs from the light.

Using UV light

If the water is cloudy, first filter it through a clean cloth, paper towel, or coffee filter. Another option is to let it settle. Then, draw off the clear water.

Follow the instructions for the UV light unit to disinfect clear water.



Filter water before using UV light to disinfect it.

Solar disinfection

In emergencies, the sun's rays can improve the quality of your water by reducing the number of germs in it. Solar disinfection does not work well in cloudy water because small particles can block germs from the light.

Using solar disinfection

If the water is cloudy, first filter it through a clean cloth, paper towel, or coffee filter. Another option is to let it settle. Then, draw off the clear water and follow the steps below.

1. Fill clean, clear plastic bottles with clear water.
2. Lay the bottles down on their side and in the sun for 6 hours (if sunny) or 2 days (if cloudy). Laying the bottles down allows the sun's rays to kill more germs in the water. You can also lay the bottles on a dark surface to help the sun's rays kill more germs.

Resources

Fact sheets



Fact Sheet: Make Water Safe During an Emergency

Available in English, Spanish, and Vietnamese.

AUG. 28, 2024

Download



Fact Sheet: Use Safe Water During an Emergency

Available in English and Spanish.

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More information

- Infant and Young Child Feeding in Emergencies Toolkit
- How to Disinfect Wells After an Emergency

- Water Disinfection for Travelers | CDC Yellow Book
- Water Treatment Options When Hiking, Camping, or Traveling
- Avoid Contaminated Water During Travel

- Natural Disasters and Severe Weather

- Emergency Disinfection of Drinking Water | U.S. EPA
- Safe Drinking Water Information | U.S. EPA